A Practical Hybrid Robust VAE-CatBoost Framework for Efficient Multiclass Network Intrusion Detection

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Abstract. Intrusion detectors that score well on random data splits often miss attack families absent from training. We present a hybrid framework built on a robust variational autoencoder (VAE) with a Student-t likelihood, masking of undefined flow statistics, and a topological loss. The VAE feeds a CatBoost classifier, and its negative log-likelihood and latent space drive two online components: a correction that relabels bursts of novel flows with the most frequent recent attack prediction, and a detector that flags unseen attack families. On CSE-CIC-IDS2018, with a time-ordered hold-out and three seeds, the framework reaches a mean per-class F1 of 0.991 ± 0.004 over the four larger classes, against 0.979 ± 0.001 for raw-feature CatBoost and 0.975 for the best of ten untuned re-run methods. The gain comes from the correction on the VAE signals, which also lifts raw-feature CatBoost to 0.990. Trained on attacks from other days, the framework with the detector reaches a Youden's J of 30.27 % on left-out days, against 6.05 % for the best of five re-run methods trained on the same days, and the detector alone flags 30.86 % of an unseen family's flows, against 4.81 % for a raw-feature detector without monotone constraints. The full pipeline needs 0.05 ms per flow on one CPU thread.

Keywords: Intrusion detection · Variational autoencoder · Student-t likelihood · CatBoost · Novelty · Unseen attack families

1 Introduction

Intrusion detection systems (IDS) that learn from network flows face skewed class distributions, heavy-tailed traffic statistics, and attack families that never appear in training. Hybrid pipelines of feature selection, autoencoders, and gradient-boosted trees report recalls near 99 % [1] and accuracies near 98 % [2] on public flow datasets, but such figures can be optimistic: random splits place flows of one attack session in both training and test data, port numbers let a model memorize attacker hosts, and accuracy hides weak rare classes [3–6]. Moreover, a closed-set classifier can output only the classes it has seen, so its detection can collapse on a new attack family [7, 8].

We keep the multi-stage hybrid design but change the role of the autoencoder. In our framework (Fig. 1), Stage 1 selects features by CatBoost SHAP values, Stage 2 encodes them with a robust variational autoencoder (VAE), and Stage 3 classifies each flow with

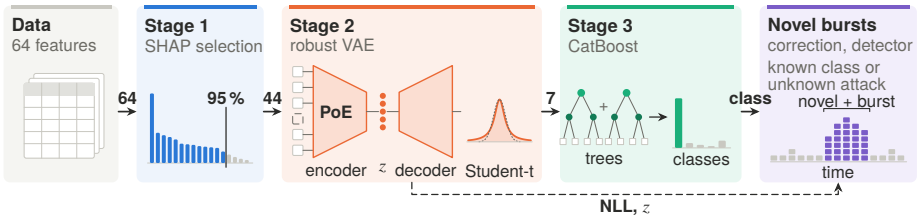


Fig. 1. Architecture. Stage 2 encodes the defined features with a product of per-feature experts (PoE); an undefined statistic (dashed) is omitted. In Stage 3 (right), the correction relabels bursts of novel flows using the VAE NLL and latent mean z, and the detector checks remaining Benign flows.

CatBoost from the VAE outputs. The VAE's reconstruction NLL also shows how unusual a flow is, and its latent space groups similar traffic; Stage 3 uses both signals in an online burst correction and an unseen-attack detector. We call Stages 1–3 without the detector *the framework*, and write *with the detector* when it is added. Our contributions are:

1. A robust VAE design for CICFlowMeter flow features that treats undefined statistics as missing entries rather than zeros, models heavy tails with a Student-t likelihood, and adds a topological loss for latent neighbourhoods; its NLL and latent space feed Stage 3 (Sect. 3).
2. A streaming correction that relabels bursts of novel flows and a monotone CatBoost detector for unseen families; both run online without host addresses (Sect. 3).
3. An evaluation on CSE-CIC-IDS2018 on a hold-out, left-out days, leave-one-attack-out folds, and a test day against ten re-run methods, with ablations and CPU timing (Sect. 4).

2 Related Work

Detection varies substantially across prior studies and can fall sharply when test attack classes are absent from training [7–10] (Table 3, bottom). Earlier hybrids pass autoencoder errors [1] or latent VAE representations [2] to tree classifiers. Streaming detectors instead maintain per-host state [11], score edge bursts between host pairs [12], or classify multi-flow windows with transformers [13]. Our streaming components track bursts in the VAE latent space and need no IP addresses.

3 Proposed Framework

3.1 Stage 1: CatBoost with SHAP Feature Selection

Stage 1 ranks the 64 candidate features by mean absolute SHAP value [14] across 20 CatBoost runs, each on a random 70 % of the features, and retains the smallest top set holding 95 % of the importance: 44 features on our data (Fig. 2, Table 1).

3.2 Stage 2: Robust Variational Autoencoder

Each part of our VAE [15] addresses a property of flow statistics.

Undefined Statistics. When CICFlowMeter cannot compute a statistic—for example, an inter-arrival time from a single packet or an idle time when no idle period exists—it writes a default value of 0 [16]. These zeros are missing entries, not measurements. In the training rows, 81.04 % of *Idle Max* values and 28.82 % of *Flow IAT Std* values are undefined. Following the CICFlowMeter source code, we mark such entries as missing ($b_j = 0$). Each feature j has its own small encoder, an expert, mapping x_j to a Gaussian $\mathcal{N}(\mu_j, \sigma_j^2)$ over the latent vector $z \in \mathbb{R}^5$. The posterior is the product of the defined-feature experts and a standard normal prior expert [17]: $\sigma^{-2} = 1 + \sum_j b_j \sigma_j^{-2}$ and $\mu = \sigma^2 \sum_j b_j \sigma_j^{-2} \mu_j$. Thus, a missing feature neither affects the posterior nor adds a likelihood term [18, 19].

Heavy Tails. Because training flow durations span six orders of magnitude and forward packet counts span four, a few extreme outliers could otherwise dominate a standard Gaussian likelihood. The decoder therefore gives each feature a Student-t likelihood with learned location, scale, and degrees of freedom [20], and negative log-likelihood ℓ_j. A soft lower bound on the log scale keeps the scale from collapsing [21].

Neighbourhoods. Stage 3's burst correction counts flows within latent grid cells. That requires similar traffic to remain close after encoding—a property the standard VAE objective does not enforce. We therefore add the 0-dimensional topological loss $\mathcal{L}_t$ of Moor et al. [22], comparing minimum spanning trees of the inputs and posterior means within each 256-flow sub-batch.

Objective. To train the experts on incomplete rows, each batch also includes a second view x_s per row that retains a random subset s of the defined features [18, 19]. The VAE minimizes

$$\mathcal{L} = \sum_{v \in \{x, x_s\}} \Big(\mathbb{E}_{q(z|v)} \big[\textstyle\sum_j m_j^v \ell_j \big] + \beta \, \mathrm{KL}\big(q(z \mid v) \,\|\, p(z)\big) \Big) + 0.1\beta \, \mathrm{KL}\big(q(z \mid x_s) \,\|\, q(z \mid x)\big) + \gamma \mathcal{L}_t, \quad (1)$$

where m_j^v is 1 for features present in view v, $p(z) = \mathcal{N}(0, I)$, and the third term aligns the two views [23]. The weight is $\gamma = 1$ (Sect. 3.4).

3.3 Stage 3: Classification, Correction and Detection

Classifier. For each flow, the VAE produces seven outputs: the five posterior means μ, the KL term, and the reconstruction NLL at $z = \mu$, summed over the 23 features defined for every flow so that the sum has the same terms across flows. A multiclass CatBoost classifier [24] with balanced class weights predicts the class from these outputs and is tuned with Optuna [25]. Since the classifier outputs only training labels, evaluations on unseen attack families treat any non-Benign prediction as a detected attack, following the binary convention of [7].

Novel-Burst Correction. A learned attack can recur in a form the classifier misses: Slowloris teardown flows appear Benign to the classifier, yet look unusual to the VAE and arrive in bursts. The correction works in three steps. First, we mark a flow as *novel* when its reconstruction NLL is above the q_n quantile of Benign NLLs on a calibration day. Second, a flow belongs to a *burst* when many flows have recently arrived in its latent cell. After dividing each latent axis by its training SD, the cell is the grid cube of side w that contains the flow's posterior mean. Because the flow records lack IP addresses, MIDAS-R [12] evaluates the latent cell in place of a host pair and flags a burst whenever its score exceeds the q_b quantile of Benign training flows. Third, a novel flow inside a burst adopts the attack class most frequently predicted by the classifier among earlier flows over the preceding W seconds of the same day, provided that this class count clears a support threshold; otherwise, its prediction stays unchanged. Since the calibration day contains no training attack classes, any attack prediction on its Benign flows is a false alarm, and the q_s quantile of these false-alarm counts sets the threshold. Every step uses only earlier flows, so the correction runs online. During the hold-out evaluation, same-day training flows remain in the stream and count as Benign predictions, so no training label adds attack support.

Unseen-Attack Detector. If a flow from an unseen attack family triggers no known-class prediction, the burst correction leaves it as Benign; a second CatBoost model, the detector, therefore examines every remaining Benign flow. Because the normal NLL level varies across capture days, its first input is the *relative NLL*: the reconstruction NLL minus the running median NLL of earlier same-day flows whose MIDAS-R score is below its 0.9 Benign training quantile. The four additional inputs are the KL term, the latent cell's MIDAS-R score, a novel-only MIDAS-R score over flows whose relative NLL exceeds its q_n quantile on calibration-day Benign flows, and the fraction of novel flows in the cell over the last W seconds. Converting each input to its percentile rank among Benign reference flows puts all days on a shared scale. Monotone constraints [26] ensure that a larger input never lowers the attack score and make it harder for the tree model to learn a narrow value range of one training family.

Its positives are labelled attacks from families absent from the VAE and classifier training labels: the training rows of the left-out class in each leave-one-attack-out fold, encoded by that fold's models, plus the attacks of day 02-28 and further capture days (Sect. 4.1). The negatives are half of the Benign flows from these days and the calibration day, with the cut-off set for 1 % false alarms on the other half. A left-out day excludes that day and its families on all days from detector training; a fold excludes its class. Any flagged flow becomes an *unknown attack*, a wrong class in the seen-class metrics.

3.4 Hyperparameter Selection

No model weight or threshold is fitted on the hold-out or test day, and the VAE stops on the 02-28 ELBO. The topological weight γ and the five correction settings are chosen with seed 3 on an inner split holding out the latest 20 % of training rows per day and class. The weight γ maximizes mean per-class F1 with the correction at its start values. The correction settings are then tuned one by one to maximize mean attack-versus-Benign F1 across three leave-one-attack-out folds of the inner split. For the detector, the NLL is

Table 1. Settings of the framework; Stage 3 values are search ranges of Optuna's TPE sampler.

Part	Settings
Stage 1	20 CatBoost runs on random 70 % feature subsets and ≤50,000 training rows per class; SHAP on 1,000 rows per class; 300 iterations, depth 6, learning rate 0.1, balanced class weights
Stage 2	all training rows; signed log, standardization on defined entries; 44 experts of 20 ReLU units, latent size 5; decoder 128 ReLU units, Student-t, soft bound log $\hat{\sigma} \geq -6$, $\nu = 1 + \text{softplus}$; $\beta = 0.5$, consistency 0.1β; second view: each defined feature kept with probability $1 - \rho$, $\rho \sim U(0, 0.7)$, and features removed that are undefined in a random other row of the batch; $\gamma \in \{0.1, 0.3, 1, 3\}$, chosen $\gamma = 1$; 256 flows per batch, distances divided by their mean; Adam $8 \cdot 10^{-3}$, batch 32,768, ≤1,000 epochs, learning rate halved after 3 flat epochs, stop after 10 epochs without a relative gain of 10^{-4} in the 02-28 ELBO
Stage 3	CatBoost, plain boosting, balanced weights; Optuna TPE, 20 trials (10 random); iterations 200–1000, depth 4–8, learning rate 0.01–0.3, L2 1–30 (both log scale); mean per-class F1 over 3 time-blocked folds of a 10 % sample per day and class (classes under 2,000 rows whole); same search for raw CatBoost
Correction	grids searched in this order from start values in brackets: $w \in \{0.1, [0.2], 0.5\}$, $q_b \in \{0.975, [0.99], 0.995\}$, $W \in \{30, [60], 120\}$ s, $q_s \in \{0.975, [0.99], 0.995\}$, $q_n \in \{0.90, 0.95, [0.975], 0.99\}$; chosen $w = 0.1$, $q_b = 0.975$, $W = 120$ s, $q_s = 0.975$, $q_n = 0.9$; latent axes divided by their training SD; MIDAS-R edge score only, decay 0.5 [12], one-minute ticks; flows of the same second are not earlier than each other
Detector	CatBoost, depth 2, 500 iterations, learning rate 0.1, monotone; rank references: relative NLL and KL of calibration-day Benign flows, stream inputs of Benign training flows; median NLL of a reservoir sample of ≤20,000 earlier flows of the day with MIDAS-R below its 0.9 Benign training quantile, refreshed every 500 flows (calibration-day median for the first 1,000); ≤200,000 negatives per day; 1 % false alarms

taken relative to either the calibration-day median NLL or the day's running median, and the inputs are chosen from five subsets; each configuration is validated by leaving out one further capture day or one fold at a time. The selected setup—the running median with all five inputs—has the highest mean detection rate among configurations with at most 2 % mean false alarms. Table 1 gives the grids and selected values.

4 Experiments and Results

4.1 Dataset and Protocol

We used the official CSE-CIC-IDS2018 CSV files [27, 28], corrected their 12-hour CICFlowMeter timestamps [16], and removed rows with non-finite values. We also removed features that could enable shortcut learning: the timestamp [4, 6], the destination port [5], 10 constant columns, and three TCP header fields that identify attacker hosts on the training days (*Init Fwd Win Byts*, *Init Bwd Win Byts*, *Fwd Seg Size Min*; related shortcuts are discussed in [4, 6]). This left 64 candidate features. We filtered exact duplicates, cross-day matches of training rows, and label conflicts, i.e., identical feature vectors with different labels. The conflicts include 184,551 FTP brute-force rows, consistent with the failed FTP attack reported in [29].

Table 2 lists the resulting splits. Test day 02-21 contains only attack families excluded from training and is our unseen-family test set. Its HOIC window includes a single-template burst of 358,621 Benign flows. Liu et al. [29] found no HOIC labelling issue beyond a time offset, so we kept these flows as Benign; only 2,125 Benign flows on that day fall outside this burst. The detector also trains on further capture days that contain at least six distinct hours, each with at least 1 % of the day's Benign flows; this rule excludes the truncated file for day 02-16. On day 02-20, we removed 797 UDP flows

Table 2. Data splits after cleaning: the hold-out contains the latest 20 % of each day and class.

Split (use)	Day(s)	Rows (class counts)
Training (Stages 1–3)	02-14, 02-15	1,039,918 (Benign 923,868; SSH 75,224; GoldenEye 33,066; Slowloris 7,730; FTP 30)
Hold-out (seen classes)	02-14, 02-15	259,983 (Benign 230,968; SSH 18,807; GoldenEye 8,267; Slowloris 1,933; FTP 8)
Validation (VAE stop, detector)	02-28	222,123 (Benign 211,538; Infiltration 10,585)
Calibration	02-23	705,316 (Benign 704,897; web attacks 419)
Test (unseen classes)	02-21	555,189 (Benign 360,746; HOIC 192,713; LOIC-UDP 1,730)
Further days (detector)	02-20, 02-22, 03-01, 03-02	6,659,249 (Benign 5,896,584; LOIC-HTTP 573,708; Bot 143,315; Infiltration 45,408; web brute force 147; XSS 64; SQL injection 23)

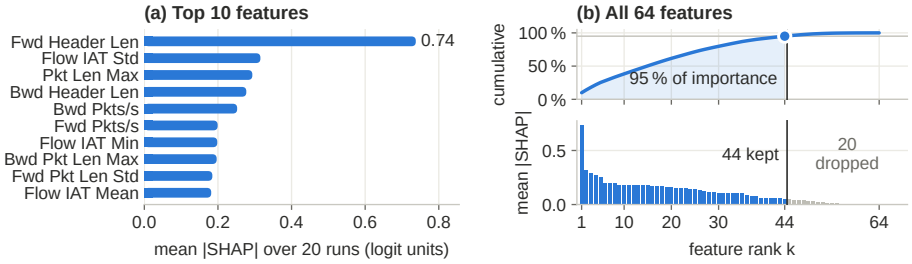


Fig. 2. Stage 1 SHAP ranking: (a) mean |SHAP| for the top 10 features over 20 runs; (b) cumulative share and importance for all 64 features (the 95 % rule keeps $k = 44$). Splitting the 20 runs into two disjoint halves gives a Spearman rank correlation of 0.963 and a top-44 Jaccard index of 0.913.

labelled LOIC-HTTP within the LOIC-UDP window, since LOIC-UDP is a test-day class.

Metrics and Runs. For seen classes, we report mean per-class F1 over classes with at least 30 hold-out rows and report the 8-row FTP class separately. For unseen families, we measure detection rate (DR), Benign false-positive rate (FPR), and Youden's J = DR − FPR. Stage 1 feature selection is run once on the main split and once per fold; all later models and thresholds are retrained for seeds 3, 4, and 5. We report the mean and standard deviation (SD) across seeds, together with a paired bootstrap comparison against raw CatBoost [30]. The VAEs and PyTorch re-run methods were trained on an RTX 5080 Laptop GPU and the other models on an Intel Core Ultra 9 275HX CPU; the code repository gives library versions, thread counts, and the command for every table: https://github.com/WinterBlossom0/IDS-VAE-and-Catboost.

Comparison Methods. Ten methods from related work—the *re-run methods*—were re-run with the same Stage 1 features of the main split or fold, the same rows, and the same seeds (Table 3), using class-balanced weights or class-balanced batches without synthetic oversampling. Each is evaluated on the hold-out, the test day, and the further capture days, and is retrained without the left-out class in each fold.

Table 3. Re-run methods (top; deviations from the source in italics; not given: absent from the source) and reported results of related work (bottom; other splits and metrics). LOAO: leave-one-attack-out

Method	Settings
Random forest [8]	scikit-learn defaults, as the source; standardization; *signed log, balanced class weights*
MLP [7]	24 hidden units (WEKA rule of the source code); *scikit-learn with early stopping, not WEKA*
XGBoost [31], LightGBM [32]	library defaults, signed log + standardization (XGBoost has no random step)
IDAE-XG [1]	as the source; top 22 features by forest rank, 20 epochs (not given); *multiclass; no upsampling*
CNN-1D [33]	as the source ($\leq$20 epochs, patience 4); *our split, stratified random split in the source*
FlowTransformer [13]	8-flow windows (source code), 2 decoder blocks, learning rate of the source grid; *multiclass*
TabM [34]	package defaults, settings of the source (patience 16); *no tuning*
FT-Transformer [35]	default configuration and optimizer (patience 16); *no tuning, no ensemble*
TabNet [36]	Table 4 of the source, patience 20; learning rate 0.02 (library default; not given); *no SMOTE*
Work	**Data; split; reported result**
Ferreira [7]	CSE-CIC-IDS2018, train 02-16, test 02-21; MLP TPR 0.898 on test attacks, 0 on LOIC-UDP
Sakr [8]	CSE-CIC-IDS2018, 9 days; random forest F1 0.957 (random split), 0.239 (LOAO mean)
Verkerken [9]	CIC-IDS2017 random split: 41/47 unknown flows found; CSE-CIC-IDS2018 Infiltration recall 78.38 %
Soltani [10]	CSE-CIC-IDS2018 packet bytes, one unknown label per run: its accuracy 35.47–72.16 %

4.2 Seen Classes: The Correction on VAE Signals Gives the Gain

We first compare the framework with a CatBoost model trained directly on the same 44 Stage 1 features (raw CatBoost) on the four seen classes with at least 30 hold-out rows (Table 4). The framework reaches a mean per-class F1 of 0.991 ± 0.004, higher than raw CatBoost (0.979 ± 0.001) and the best re-run method (0.975). For seeds 3, 4, and 5, its margin over raw CatBoost is 0.008–0.017, with 95 % paired bootstrap intervals [0.006, 0.010], [0.015, 0.020], [0.009, 0.013]. Turning on the detector lowers F1 slightly to 0.990 ± 0.004 because it flags about 1 % of Benign hold-out flows as unknown attacks. Table 5 shows that the gain comes from Slowloris; raw CatBoost does better on the 8-row FTP class and therefore over all five classes (0.979 versus 0.940). The ablations in Table 4 locate the source of this gain. Without the correction, the VAE-fed classifier is below raw CatBoost (0.965 versus 0.979). The correction raises its Slowloris recall from 0.79 (not tabulated) to 0.98 and also lifts raw CatBoost to 0.990, which matches the framework within the seed spread. Even so, raw CatBoost with the correction still must run the VAE to compute NLLs and latent cells, relabels more Benign flows, and is slower than the framework. The gain therefore comes from the correction using VAE signals, not from the VAE serving as a feature compressor. Consistent with this, a Gaussian likelihood, removal of the topological loss, or use of latent means alone changes F1 by at most 0.003 with the correction, within the seed spread; without the correction, the topological loss lowers F1. On unseen families, a detector on a VAE without the topological loss reaches a left-out-day J of 26.17 %, versus 26.37 % with it, when both train only on the further days; the loss therefore does not change detection either.

Using LightGBM or XGBoost in Stage 3 gives a similar F1 but lower balanced accuracy. Concatenating the 44 raw features with the seven VAE outputs still trails raw CatBoost before correction (0.973 versus 0.979), ties the framework after correction (0.991), and removes the FTP false alarms (FTP F1 0.98 versus 0.74).

Table 4. Hold-out mean per-class F1 (classes ≥30 rows) per seed and mean ± SD, balanced accuracy (all classes), Benign hold-out flows relabelled (sum over seeds; with the detector: by the correction and the detector), and CPU ms per flow (batch 1, one thread, seed 3). With the correction, the framework has a higher mean F1 than raw CatBoost and each re-run method.

Configuration	Seed 3	Seed 4	Seed 5	Mean F1	Bal. acc.	Relab.	ms/flow
Ours (with the correction)	0.9883	0.9949	0.9895	0.9909 ± 0.0036	0.9947 ± 0.0018	487	0.044
Without the correction	0.9583	0.9700	0.9667	0.9650 ± 0.0060	0.9569 ± 0.0051	–	–
Without the topological loss	0.9880	0.9931	0.9906	0.9905 ± 0.0026	0.9865 ± 0.0153	405	–
No topological loss or corr.	0.9742	0.9655	0.9772	0.9723 ± 0.0060	0.9590 ± 0.0255	–	–
Gaussian likelihood	0.9876	0.9908	0.9864	0.9883 ± 0.0023	0.9940 ± 0.0008	583	–
Latent mean only (5 inputs)	0.9883	0.9913	0.9875	0.9890 ± 0.0020	0.9946 ± 0.0013	614	–
LightGBM in Stage 3	0.9905	0.9935	0.9900	0.9913 ± 0.0019	0.9767 ± 0.0284	671	0.16
XGBoost in Stage 3	0.9910	0.9954	0.9905	0.9923 ± 0.0027	0.9441 ± 0.0015	614	0.058
Raw + VAE inputs (51) + corr.	0.9902	0.9893	0.9934	0.9910 ± 0.0022	0.9950 ± 0.0007	1209	–
Raw + VAE inputs (51)	0.9781	0.9634	0.9788	0.9734 ± 0.0087	0.9673 ± 0.0085	–	–
Raw CatBoost + correction	0.9911	0.9896	0.9907	0.9904 ± 0.0008	0.9955 ± 0.0003	1377	0.057
Raw CatBoost (44 features)	0.9800	0.9776	0.9785	0.9787 ± 0.0012	0.9739 ± 0.0014	–	0.018
Ours + detector	0.9868	0.9937	0.9883	0.9896 ± 0.0036	–	7674	0.050
Re-run methods (Table 3)							
Random forest [8]	0.9642	0.9643	0.9643	0.9643 ± 0.00004	0.9042 ± 0.00004	–	0.0040
MLP [7]	0.9781	0.9555	0.9600	0.9645 ± 0.0120	0.9628 ± 0.0148	–	0.0031
XGBoost [31]	0.9640	0.9640	0.9640	0.9640 ± 0	0.9042 ± 0	–	0.0053
LightGBM [32]	0.9655	0.9677	0.9629	0.9654 ± 0.0024	0.9549 ± 0.0002	–	0.0059
IDAE-XG [1]	0.6959	0.6423	0.6425	0.6602 ± 0.0309	0.6621 ± 0.0215	–	0.016
CNN-1D [33]	0.9742	0.9780	0.9734	0.9752 ± 0.0024	0.9740 ± 0.0032	–	0.039
FlowTransformer [13]	0.8276	0.8279	0.8402	0.8319 ± 0.0072	0.8827 ± 0.0189	–	0.058
TabM [34]	0.9493	0.9510	0.9502	0.9502 ± 0.0008	0.9542 ± 0.0001	–	0.39
FT-Transformer [35]	0.9608	0.9606	0.9626	0.9613 ± 0.0011	0.9546 ± 0.0014	–	0.74
TabNet [36]	0.9546	0.9531	0.9632	0.9569 ± 0.0054	0.9594 ± 0.0058	–	0.098

Inference Time. On a single CPU thread, end-to-end processing with the detector takes 0.050 ms per flow, compared to 0.018 ms for raw CatBoost: the fused VAE and CatBoost graph accounts for 0.044 ms (0.046 ms as two separate graphs), the correction takes 0.00015 ms, and the detector takes 0.0058 ms. We timed each pipeline with a time in Table 4 as ONNX Runtime 1.31 graphs and C++ code on one thread of an Intel Core Ultra 9 275HX, one flow per call from the extracted features to the final label, over the 100,000 consecutive flows in the middle of day 02-21 (98,562 Benign burst and 1,438 attack flows). Each time is the median of 4 passes; a full repeat gave the same order of the 16 pipelines and a framework time 1.3 % higher.

Table 6 measures F1 and cost with the k most important features of Stage 1 (seed 3; $k = 64$: all 79 columns minus the 15 dropped); for $k > 10$, our F1 stays within 0.9870–0.9908, and $k \in \{23, 40, 64\}$ score above the $k = 44$ of the 95 % rule.

4.3 Unseen Attack Families

Table 7 tests the detector alone on left-out days and the framework with the detector in the leave-one-attack-out folds. The detector mainly catches families whose flows arrive as bursts of novel flows. For LOIC-HTTP, it flags 99.01 % of request flows but only 0.28 % of unanswered attempts, while Infiltration stays mostly undetected. The VAE-based inputs are key: a CatBoost detector on the 44 raw features instead of its five inputs, trained on the same rows with the same cut-off but without monotone constraints, flags

Table 5. Per-class hold-out results, mean over seeds 3–5, of the framework and raw CatBoost (top), and the confusion matrix of the framework, seed 3 (bottom; rows: true class)

		Ours (with correction)			Raw CatBoost (44 features)		
Class	Rows	Precision	Recall	F1	Precision	Recall	F1
Benign	230,968	0.9998	0.9978	0.9988	0.9989	0.9981	0.9985
DoS GoldenEye	8,267	0.9964	0.9992	0.9978	0.9983	0.9998	0.9990
DoS Slowloris	1,933	0.9801	0.9767	0.9784	0.9929	0.8724	0.9288
FTP-BruteForce	8	0.6218	1.0000	0.7360	0.9630	1.0000	0.9804
SSH-BruteForce	18,807	0.9775	1.0000	0.9886	0.9777	0.9995	0.9884
Balanced accuracy		0.9947 ± 0.0018			0.9739 ± 0.0014		
Macro ROC-AUC (one vs rest, classifier)		0.9993 ± 0.0005			0.9995 ± 0.0001		
Macro PR-AUC (one vs rest, classifier)		0.9689 ± 0.0219			0.9924 ± 0.0011		

True \ predicted	Benign	GoldenEye	Slowloris	FTP	SSH
Benign	230,374	34	56	0	504
DoS GoldenEye	8	8,259	0	0	0
DoS Slowloris	59	0	1,874	0	0
FTP-BruteForce	0	0	0	8	0
SSH-BruteForce	1	0	0	0	18,806

Table 6. Feature-count ablation (seed 3): hold-out mean per-class F1, VAE training time in s (one VAE at a time on the GPU) and inference time in ms per flow (batch 1, one thread, no detector, as Table 4)

	F1				ms per flow	
k	Ours	Raw	Raw + corr.	VAE (s)	Ours	Raw
10	0.9738	0.9815	0.9881	141	0.046	0.017
15	0.9877	0.9792	0.9886	158	0.072	0.019
23	0.9908	0.9782	0.9895	273	0.051	0.015
30	0.9870	0.9799	0.9908	357	0.054	0.024
40	0.9902	0.9813	0.9911	302	0.061	0.019
44	0.9883	0.9800	0.9911	317	0.044	0.018
64	0.9908	0.9805	0.9906	392	0.057	0.016

on average only 4.17 % (depth 6) and 4.81 % (depth 2) of each left-out family's flows, versus 30.86 % for ours (mean of Table 7a), showing that the labelled further days and tree depth alone do not explain the detection.

Table 8 compares all methods by Youden's J. Trained on days 02-14 and 02-15 only, our classifier scores like most re-run methods (J 0.57 % on left-out days); with the detector, also trained on the further days, the framework reaches 30.27 % and 78.29 % in the folds. Since the detector configuration was selected on these days and folds (Sect. 3.4), only the test day is unused by that choice. The test day shows the limit of the burst signal: 99.4 % of its Benign flows belong to one burst inside the HOIC window. The framework detects 81.35 % of LOIC-UDP flows, but also flags 66.77 % of this Benign burst, which receives higher detector scores than the HOIC flows. In the rows marked $^{+}$ in Table 8, the five non-deep re-run methods also train on the further days except the evaluated one, with each attack flow labelled by family, and flag flows by the detector's cut-off rule; Isolation Forest [37], an open-set detector, trains on their Benign flows only. Given the same days, the best reach J 6.05 % on left-out days (LightGBM) and 34.35 % in the folds (MLP), versus 30.27 % and 78.29 %.

Table 7. Attack families absent from all training labels, in %, mean ± SD over seeds 3–5: (a) left-out days, DR and Benign FPR of the detector alone; (b) leave-one-attack-out folds, DR on days 02-14 and 02-15 after each step, and hold-out Benign FPR; with raw CatBoost as classifier (same correction and detector), the mean final DR is 0.01 to 1.67 points higher per fold.

(a) Left-out days

Day	Family	DR	FPR
02-28	Infiltration	2.34±1.83	2.42±0.23
02-20	LOIC-HTTP	50.06±0.12	0.91±0.16
02-22	Web brute force	30.84±21.21	0.61±0.06
	SQL injection	13.04±7.53	
	XSS	48.44±17.61	
03-01	Infiltration	3.03±0.71	0.68±0.01
03-02	Bot	68.24±2.06	0.68±0.04

(b) Leave-one-attack-out folds

Removed class	Classifier	+ corr.	+ det.	FPR
SSH-BruteForce	0.0007±0.001	0.0007±0.001	95.16±5.02	2.19±1.21
DoS GoldenEye	0.11±0.01	11.96±1.40	58.19±19.27	1.68±0.92
DoS Slowloris	53.01±13.28	71.09±2.01	87.43±8.32	2.03±0.97

Table 8. Unseen attack families, in %, mean over seeds 3–5: Youden's J is the detection rate (DR) minus the Benign false-positive rate (FPR); a re-run method flags a flow whose argmax class is not Benign. $^{+}$: trained also on the further days and flagged at the detector's cut-off. Left-out days: mean over day and family; folds: mean over the three folds. Test day 02-21: DR, Benign FPR outside and inside the burst, and J against all Benign flows.

	J		Test day					
	Left-out days	Folds	DR UDP	DR HOIC	FPR out	FPR burst	J UDP	J HOIC
Ours + detector	30.27	78.29	81.35	0.01	15.42	66.77	14.88	−66.46
Random forest [8]	0.03	15.81	0	0	0.27	0.0003	−0.002	−0.002
MLP [7]	1.60	18.99	0	0	0.72	4.05	−4.03	−4.03
XGBoost [31]	−0.01	23.90	0	0	0.38	0	−0.002	−0.002
LightGBM [32]	−0.01	5.36	0.15	0	0.35	0.00009	0.15	−0.002
IDAE-XG [1]	24.57	36.52	99.81	32.17	68.17	67.64	32.16	−35.48
CNN-1D [33]	0.60	16.56	0.02	0	0.49	20.57	−20.43	−20.45
FlowTransformer [13]	8.33	8.02	93.29	17.32	38.35	68.57	24.90	−51.07
TabM [34]	0.02	16.84	63.91	0	1.36	0.57	63.34	−0.57
FT-Transformer [35]	0.31	20.85	6.61	0	0.64	0.94	5.67	−0.94
TabNet [36]	0.07	16.94	0	0.23	0.53	25.01	−24.87	−24.64
Random forest^{+}	0.16	−0.01	0	0.43	0.05	0	−0.0003	0.43
MLP^{+}	1.18	34.35	99.83	0.0002	1.38	27.43	72.55	−27.27
XGBoost^{+}	0.98	24.08	0	0.11	0.45	0	−0.003	0.11
LightGBM^{+}	6.05	22.96	0.50	0.33	0.28	0	0.50	0.33
IDAE-XG^{+}	−0.25	0.68	46.22	0.34	1.22	22.01	24.34	−21.54
Isolation Forest^{+} [37]	0.36	−0.65	66.69	0	33.44	0.01	66.48	−0.21

5 Limits

Unusual Benign bursts can trigger false alarms, attacks without bursts give no signal, and the correction assumes one dominant attack class per time window. The evidence comes from three seeds on one dataset with known errors and label faults [4, 29]. The deep re-run methods (CNN-1D, FlowTransformer, TabM, FT-Transformer, TabNet) were not given the further days, Kitsune [11] was not run, re-run methods use defaults without the correction, and missing-entry handling was not ablated.

6 Conclusion

We proposed a hybrid VAE-CatBoost framework in which a robust VAE serves both a classifier and two streaming components. On CSE-CIC-IDS2018, the correction lifts the mean per-class F1 of the four larger classes above raw CatBoost and ten untuned re-run methods, and the framework with the detector reaches a Youden's J of 30.27 % on left-out days, versus 6.05 % for the best untuned re-run method given the same days. A large Benign burst on the test day defeats the detector; separating such bursts from attacks remains future work.

Use of AI Tools. Claude Code was used for the implementation of the code and for finding bugs.

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